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Similarity Measure Based on Incremental Warping Window for Time Series Data Mining

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ABSTRACT A similarity measure is one of the most important tasks in the fields of time series data mining. Its quality often affects the efficiency and effectiveness of the related algorithms that need to measure the similarity between two time series in advance. Dynamic time warping is one of the most robust methods to compare one time series with another based on warping alignments. In this paper, the design of an incremental warping window is used to improve the performance of dynamic time warping. The incremental warping window is changeable for various time series with different lengths. Moreover, the improved dynamic time warping based on the novel window considers the recent alignments as much as possible, which indicates that the proposed method concentrates on more information of the recent data points than that of the previous data points. In addition, it is suitable for online similarity measure between data stream. The experimental evaluation shows that the proposed method is effective and efficient for time series mining.

INDEX TERMS Dynamic time warping, similarity measure, time series data mining, incremental warping window, classification.

I. INTRODUCTION

Time series is one of the most popular data in the fields of data mining. Time series data mining is also one of the most challenged directions [1], [2] and attracts much attention [3]–[5]. The main tasks of data mining in time series datasets include clustering [1], [5], [6], classification [7], [8], fault detection [3], pattern recognition [9] and prediction [10], [11]. A common and important task in the fields of time series data mining is similarity measure between two objects [12], which compares one time series with another. Sometimes, we often call it distance measure. In some domains, a simple similarity measure like Euclidean distance is sufficed. However, it is often sensitive for the abnormal data points and requires that the lengths of the compared time series are equal.

In most cases, two compared time series have the approximately same shapes that do not line up in time axis. In other word, the approximately same shapes are on the two time series in different time. We hope to warp the time axis of one or both to align them so that good similarity measure between two time series can be achieved. Dynamic time warping (DTW) [13]–[16] is a such technique for efficiently measure many good warping alignments. It is often applied to gesture and status recognition [17], speech processing [18], [19], image recognition [20], [21], medicine [22], [23] and other fields [24]–[27].

Beside the applications of DTW, many researches also concentrate on measure quality and time complexity. The measure quality of DTW impacts on the quality of algorithms used in the fields of time series data mining [3], [6], [9], [28], such as classification, clustering, pattern recognition, motifs and abnormal discovery. Keogh and Pazzani [29] proposed an improved version of DTW based on derivative, that is derivative dynamic time warping (DDTW). It could reduce the pathological singularity. Fu *et al.* [30] proposed an uniform scaling to allow global scaling of time series, and discussed its utility and effectiveness on a wide range of problems in medicine, industry and entertainment. Jeong *et al.* [31] proposed weighted dynamic time warping for time series classification, which is a penalty-based DTW that can achieve improved accuracy of time series classification and clustering problems. In some work, Li [32] proposed an online and dynamic time warping based on a suitable connection of two warping paths that derive from the two local window. It can improve the performance of the classification for time series. Due to the square time complexity of DTW, it needs to reduce the time consumption. Since the high time complexity is caused by the construction of a cost matrix and the search of the optimal warping path in the cost matrix, much attention concentrates on reducing the scope to find the optimal warping path in the cost matrix. Two extensions to DTW that significantly speed up the DTW calculation is a technique based on global constraints, the Sakoe-Chiba Band [33] and Itakura Parallelogram [34]. Moreover, some lower bound functions [35]–[38] based on constraint band are often applied to time series similarity search and index.

Due to the importance of DTW based on the global constraint Sakoe-Chiba Band for similarity measure between time series, the main motivation of this work is to propose an improved constraint for DTW, which not only can extend the traditional constraint but also can be used to measure similarity between any two stream time series. Moreover, the scope of the constraint is changeable with the increasing length of the compared time series. It concentrates on the local constraint and more information of the recent data points than that of the previous (historical) data points. In addition, the experimental results demonstrate that the proposed method outperforms the traditional one for similarity measure between time series.

II. DYNAMIC TIME WARPING

Dynamic time warping (DTW) [29] is a method used to warping measure the similarity between two time series. It allows the approximately same shapes of different time series to align each other, which makes DTW reflect the true distance between the two time series. In addition, it is unsensitive to the abnormal data points and can deal with the similarity between two time series with different lengths. Due to the merits of DTW [35], [38], [39], it is often applied to the field of time series data mining such as gesture recognition, speech processing, image recognition and medicine expert systems.

Give two time series $X = \{x_1, x_2, \cdots, x_m\}$ and $Y = \{y_1, y_2, \cdots, y_n\}$, where it is often $n \neq m$, n and m respectively denote the lengths of the two time series. DTW minimizes the distance between two time series X and Y by selecting the best warping path P_0 from a set of warping path set P, that is $P_0 \in P$. Any warping path $P_i \in P$ can be constructed from a cost matrix R of which the cell element $R(i, j)$ is regarded as the sum of the distance $D(i, j)$ plus the minimum of the cumulative distance of the three adjacent elements of $R(i, j)$, that is

$$R(i,j) = D(i,j) + min \begin{cases} R(i, j-1) \\ R(i-1, j-1) \\ R(i-1, j) \end{cases} \quad (1)$$

$D(i, j)$ represents the distance between data points x_i and y_j, that is often $D(i, j) = (x_i - y_j)^2$. In this way, the best warping path P_0 can be constructed by

$$\begin{aligned} DTW(X, Y) &= R(m, n) \\ &= \min_{P_0} \sum_{k=1}^{K} D(p_k), \end{aligned} \quad (2)$$

where p_k is the element of the best warping path P_0 and denotes the matched times (i_k, j_k), which must satisfies three constraint conditions(Boundary, Continuity and Monotonic) [29]. As shown in 1, the two time series are compared by aligning the approximately same shapes in different times.

It is well known that DTW needs to construct a cost matrix R of size $m \times n$, which means that DTW should cost the time and space complexity $O(mn)$. When the lengths of the compared time series is large, DTW becomes an inefficient method to measure the similarity. There are some methods used to speed up the calculation of DTW. Two of the earliest methods are the Global constraints, they are the Sakoe-Chiba Band [33] and the Itakura Parallelogram [34]. They are used to limit the scope of the cells surrounding the optimal warping path in the cost matrix. In especial, the Sakoe-Chiba Band is widely used in the community of similarity search for time series [35], [38]. As shown in Figure.2, the best warping path in red is only in the scope of the green cells that constructed by a warping window l. In this way, DTW based on the warping window can be faster than the traditional one.

Some researches [40] pointed out that the cells in the cost matrix produced by a warping window l are enough to make a good distance value for similarity search and index. In most cases, the warping window l is equal to the r percent of the minimum length of the two compared time series. That is $l = \lfloor min(m, n) * r \rfloor$, where r is often set to be equal or less than 0.1 [40].

In addition, some modifications of DTW based on Sakoe-Chiba Band that speed up the calculation of DTW are the lower bounding functions based on warping windows. LB_Keogh [35] is regarded as the Euclidean distance between any parts of the candidate matching sequence not falling within an envelope and the nearest corresponding section of the envelope. It is often used to fast remove the dissimilar objects when used to search or index the similar time series. Some extensions [41]–[43] based on principle of *LB_Keogh* were proposed to improve the applicable performance of the Sakoe-Chiba Band for DTW.

III. DTW BASED ON INCREMENTAL WARPING WINDOW

A. WORK MOTIVATION

DTW based on Sakoe-Chiba Band (here it is abbreviated to SDTW) is more faster to execute the procedure than the original one and also can obtain the approximately true distance that makes a good result of time series data mining, including classification, clustering and similarity search. However, the length of the warping window in Sakoe-Chiba Band should be preset before executing DTW and is decided by the length of the compared time series. In other words, the length l of the warping windows is set to be $\lfloor r * min(m, n) \rfloor$, where m and n are respectively the lengths of the compared time series, r is the percentage of the maximum. It has some disadvantages as follows.

(1) A long time series has the large length, which makes the warping widow is big although the percentage r is often set to be 0.1. It means that the long compared time series makes the length of the warping windows is large, which causes many cells in the scope of cost matrix used to find the optimal warping path.

(2) The length of the warping window based on the Sakoe-Chiba Band is a const value. It means that different time points have the same size of the matching area. Moreover, it is a global constraint for the search scope of finding the optimal warping path and does not consider the local warping constraint. In particular, the information of the recent data points and the previous ones are treated equally without discrimination.

(3) The fixed length of the warping window based on the Sakoe-Chiba Band is suitable for the compared time series with an invariable length. It means that DTW based on the Sakoe-Chiba Band is not robust for the stream time series with the changeable lengths.

Due to the above mentioned reasons, we propose a new method to constrain the warping window for DTW. The work motivation includes four aspects. Firstly, an incremental warping window used in DTW are proposed to improve the performance of DTW based on Sakoe-Chiba Band. Secondly, The length of proposed warping window is changeable with the length of the compared time series. Thirdly, DTW based on incremental warping window is suitable for similarity measure between the stream time series. In addition, DTW based on incremental warping window concentrates on the more information of the recent data points than that of the previous data points.

B. INCREMENTAL WARPING WINDOW

Given two stream time series $X^{m_0} = \{x_1, x_2, \cdots, x_{m_0}\}$ and $Y^{n_0} = \{y_1, y_2, \cdots, y_{n_0}\}$, where m_0 and n_0 are respectively the lengths of the two time series. An incremental warping window (IWW) can be defined as a changeable local warping window according to the lengths of the two time series. Suppose the lengths of two increased stream time series X^{m_i} and Y^{n_i} are m_i and n_i, where $i = 2, 3, \cdots, m_i > m_0$ and $n_i > n_0$. Formally, the length l_i of the incremental warping window for the compared stream time series X^{m_i} and Y^{n_i} is

$$l_i = \lfloor min(m_i, n_i) * r \rfloor, \tag{3}$$

where r is a percentage value and is often set to be 0.1 [40]. The lengths m_i and n_i are changed when the compared stream time series add some coming data points, which causes the length of the warping window is always variable. Like Sakoe-Chiba Band, the incremental warping window seems to be easy to design. (3) tells us that the length can be set when the lengths of the two compared stream time series are known. However, in most cases the lengths of the two steam time series are unknown. In other words, we often do not know the beginning of the stream time series.

Suppose there are two different phases i and j of the two stream time series, where $j > i$. Thus the corresponding pairs of the stream time series at the different phases are (X^{m_i}, Y^{n_i}) and (X^{m_j}, Y^{n_j}). Moreover, if the time gaps of the two pairs are w and h, that is

$$\begin{cases} w = m_j - m_i \\ h = n_j - n_i \end{cases} \tag{4}$$

Again, suppose the incremental warping window is obtained at the phases i, which can be computed by (3). Now we want to obtained the length of the new incremental warping window at the phases j. It is easy to calculate the length by

$$l_j = \lfloor min(m_i + w, n_i + h) * r \rfloor. \tag{5}$$

Combining with (3), (4) and (5), the corresponding incremental warping window for a changeable time series can be obtained. In particular, the distance between two time series X and Y with different lengths m and n can be computed by DTW based on incremental warping window. The length of the incremental warping window can be decided by

$$l_j = \min \begin{cases} max(b, \lfloor (m_i + w) * r \rfloor) \\ max(b, \lfloor (n_i + h) * r \rfloor), \end{cases} \tag{6}$$

where b is the initial length of the incremental warping window and is equal or greater than 0, that is $r \geq 0$.

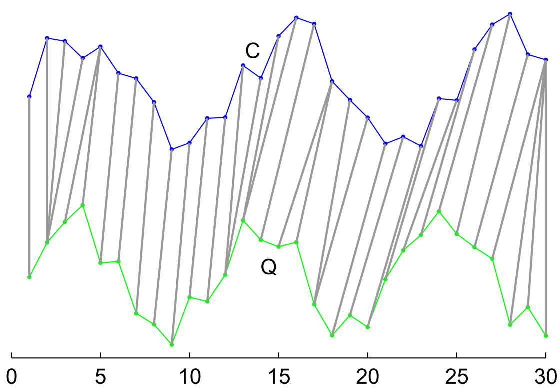

FIGURE 1. The warping alignment between two time series by DTW.

For the two time series Q and C as shown in Figure.1, the scope of the cells used to produce the warping path based on IWW is shown in Figure.3. It is easy to find that the best warping path is surrounded by the cells. Moreover, the lengths of the warping windows at different time are various. We initialize b to be 0 as shown Figure.3, which means that the beginning length of the warping window is 0. When b is set to 1, the search scope of the cells are wider than before and surround the best warping path. Actually, when b is equal to 3, the warping window of the two compared time series are degenerated into the Sakoe-Chiba Band as shown in Figure.2.

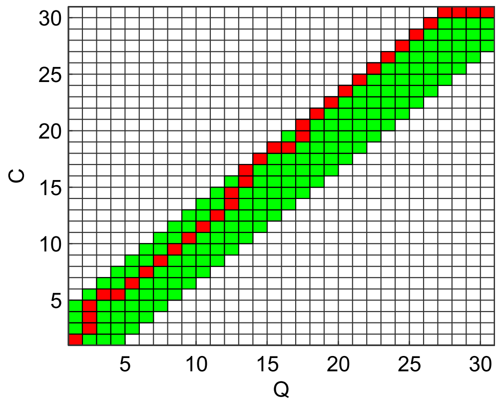

FIGURE 2. **The cells in the cost matrix are used to produce the warping path by DTW.**

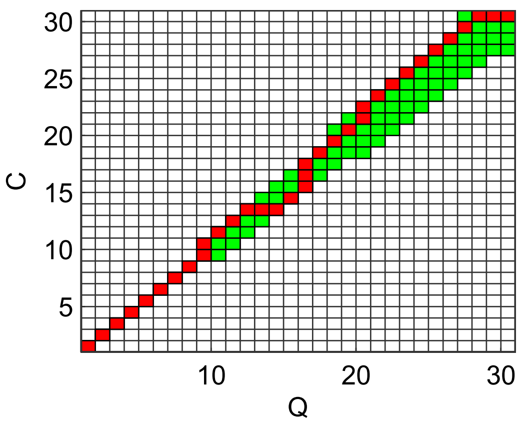

FIGURE 3. **The cells in the cost matrix are used to produce the warping path based on incremental warping window.**

C. ALGORITHM OF DTW BASED ON IWW

As shown in Figure.3, the optimal warping path based on incremental warping window can be obtained. Moreover, the farther the time is, the less the length of the incremental warping window is. In other words, DTW based on incremental warping window considers more matching information of the recent data points than that of the previous data points. It means that the alignments between two subsequences close to the newest time are better than historical ones. It also indicates that the distance contribution of the historical data points to measure the similarity is less important than that of the recent data points. Actually, it is interesting to measure the similarity between two long time series. The reason is that the recent information is much more important and useful than the historical information to make decision. The proposed version of DTW is called as incremental dynamic time warping (IDTW).

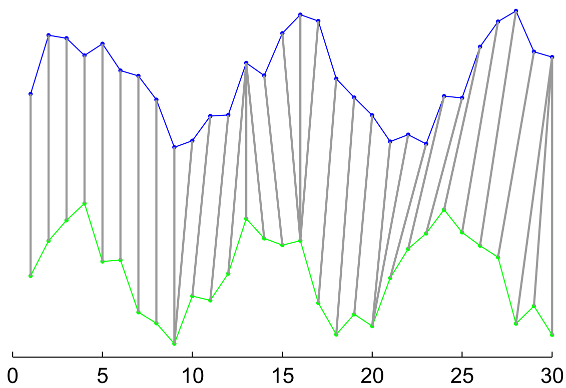

FIGURE 4. **The warping path produced by IDTW for time series of the length 60.**

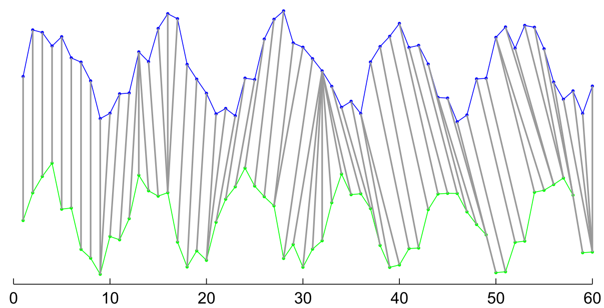

FIGURE 5. **The incremental warping path produced by IDTW for the time series shown in subfigure (a) of which the length is increased to 60.**

For the above instance, the warping alignments between two stream time series are produced by IDTW as shown in Figure.4. In this case, b in (5) is 0. Moreover, w and h are set to 1, which indicates that the length of the coming subsequences is 1. It simulates the work of stream time series. In Figure.4 and Figure.5, the historical data points of which the time are less than 10 are aligned to each other at same time. With the increasing length of the compared stream time series, the alignments between them are more and more better, which can match the subsequences with the approximately same shapes to each other. As shown in Figure.5, the recent alignments from time point 40 to 60 reflect the changeable information of the approximately same shapes. It means that the recent subsequences can be aligned better than the historical ones by IDTW.

We can modify the algorithm of DTW to add the incremental constraint and design the algorithm of IDTW as illustrated in algorithm 1.

The parameters X and Y in the function of IDTW denote the historical subsequences and X_w and Y_h represent the coming subsequences. Line 1 initializes the value of b and adds one zero element as the beginning point of the new series X and Y respectively. It also connects the coming subsequence. Line 2 computes the length of the incremental warping window as shown in (6). Lines 4, 5, 11 and line 12 respectively compute the beginning position of every row and column cells in R. Moreover, the column cells in green as shown in Figure.3 can be calculated from line 7 to line 8.

Algorithm 1 $[dist, R, X, Y, m, n]$=IDTW(X,Y,X_w, Y_h,R,m,n,r)

1: $X = 0 \bigcup X \bigcup X_w$; $Y = 0 \bigcup Y \bigcup Y_h$; $b = 0$
2: $l = \min(max(b, \lfloor (m + w) * r \rfloor), max(b, \lfloor (n + h) * r \rfloor))$;
3: **for** $j = m + 2$ to $j = m + w + 1$ **do**
4: $\quad begin = min(max(2, j - l), n + h)$;
5: $\quad R(begin, j) = inf$
6: $\quad$ **for** $i = begin + 1$ to $j = n + 1$ **do**
7: $\quad\quad R(i, j) = (q_i - c_j)^2 + min(R(i-1, j), R(i, j-1), R(i-1, j-1))$;
8: $\quad$ **end for**
9: **end for**
10: **for** $i = n + 2$ to $i = n + h + 1$ **do**
11: $\quad begin = min(max(2, i - l), m + w)$;
12: $\quad R(i, begin) = inf$
13: $\quad$ **for** $j = begin + 1 : min(i + l, m + w + 1)$ **do**
14: $\quad\quad R(i, j) = (q_i - c_j)^2 + min(R(i-1, j), R(i, j-1), R(i-1, j-1))$;
15: $\quad$ **end for**
16: **end for**
17: **if** $j \neq m + w + 1$ **then**
18: $\quad i = n + h + 1$;
19: $\quad$ **for** $k = j : m + w + 1$ **do**
20: $\quad\quad R(i, k) = (q_i - c_k)^2 + min(R(i - 1, k), R(i, k - 1), R(i - 1, k - 1))$;
21: $\quad$ **end for**
22: **end if**
23: $dist = R(n+h+1, m+w+1)$; $m = m+w$; $n = n+h$;
24: $X(1) = []$; $Y(1) = []$

Algorithm 2 $[dist, path]$=Equal_Length_IDTW(X, Y,r)

1: Initialize some variables such as $Q = []$,$C = []$, $m = n = 0$ and $R = []$;
2: **for** $i = 1$ to $i = length(X)$ **do**
3: $\quad [dist, R, path, Q, C, n, m]$=NIDTW($Q$,$C$,$X(i)$, $Y(i)$,R,r,n,m);
4: **end for**

Similarly, the row ones can be computed from line 13 to 15. Line 17 to 22 are used to ensure that the green cells can arrive to the last cell $(m + w + h, n + h + 1)$. Line 23 retrieves the distance between the two newest stream time series and their lengths. The last line is used to remove the first element which is zero added in line 1.

For any two time series $X = \{x_1, x_2, \cdots, x_m\}$ and $Y = \{y_1, y_2, \cdots, y_n\}$, the algorithm of IDTW can be executed to get the best warping path and the corresponding distance value. The way based on IDTW is shown in Algorithm 2 and is denoted as Equal_Length_IDTW.

Here, we only give the algorithm 2 that are suitable to measure the similarity between two time series with equal length by IDTW. It can be properly modified for comparing the time series with different lengths. In the algorithms 1 and 2, it is easy to know that the proposed method can handle the similarity of the stream time series. It is suitable to online compute the distance between any two time series. To speed up the calculation of algorithm 1, two cost vectors including the row one and the column one replacing the cost matrix can be considered to ingeniously and repeatedly update the previous row and column cost cells, which is applied to the experimental evaluation of IDTW.

IV. EXPERIMENTAL EVALUATION

In this section, The traditional DTW based on Sakoe-Chiba Band (here named as SDTW) and the proposed one based incremental warping window (IDTW) are used to perform on UCR time series datasets. The nearest neighbor classification are applied to evaluation of the methods for their performance. Two aspects of the comparison including accuracy and efficiency are designed to verify their performance.

A. TIME SERIES DATASETS

In the section of experimental evaluation, there are 20 data sets carried out by experiments. The datasets originate from different domains, such as medicine, robotics, handwriting recognition and so on. They are composed of different univariate time series having different number of classes, lengths, training dataset size and testing dataset size. Each dataset is composed of two subsets, the training set and the testing set. The detailed information of all the datasets carried out by the experiments is shown in Table 1.

TABLE 1. Information of UCR datasets.

ID	Name	Class number	Train size	Test size	Length
1	Adiac	37	390	391	176
2	Beef	5	30	30	470
3	CBF	3	30	900	128
4	Coffee	2	28	28	286
5	ECG200	2	100	100	96
6	FISH	7	175	175	463
7	FaceAll	14	560	1,690	131
8	FaceFour	4	24	88	350
9	Gun_Point	2	50	150	150
10	Lighting2	2	60	61	637
11	Lighting7	7	70	73	319
12	OSULeaf	6	200	242	427
13	OliveOil	4	30	30	570
14	SwedishLeaf	15	500	625	128
15	Trace	4	100	100	275
16	Two_Patterns	4	1,000	4,000	128
17	Synthetic_Control	6	300	300	60
18	Wafer	2	1,000	6,174	152
19	50Words	50	450	455	270
20	Yoga	2	300	3,000	426

B. CLASSIFICATION

The nearest neighbor classification is the best classifier used the testify the performance of the distance functions. It can reflects the quality of similarity measure. In this section, we used it to perform on all the datasets in Table 1. Let every

TABLE 2. The error rates of classification are produced by six methods.

ID	Name	DTW	SDTW	IDTW$_{mean}$	IDTW$_{min}$	ED	SpADe	$2DD_{LCSS}$
1	Adiac	0.396	0.379	0.396	0.386	0.389	0.438	0.575
2	Beef	0.500	0.500	0.460	0.433	0.333	0.500	0.467
3	CBF	0.003	0.002	0.028	0.009	0.148	0.044	0.012
4	Coffee	0.179	0.250	0.257	0.179	0	0.185	0.107
5	ECG200	0.230	0.210	0.146	0.110	0.12	0.256	0.13
6	FISH	0.167	0.171	0.183	0.171	0.217	0.150	0.057
7	FaceAll	0.192	0.119	0.192	0.154	0.231	0.315	0.085
8	FaceFour	0.170	0.102	0.124	0.114	0.216	0.250	0.182
9	Gun_Point	0.093	0.120	0.042	0.040	0.087,	0.007	0.033
10	Lighting2	0.131	0.131	0.131	0.131	0.246	0.272	0.18
11	Lighting7	0.274	0.329	0.287	0.287	0.425	0.167	0.452
12	OSULeaf	0.409	0.392	0.398	0.388	0.479	0.212	0.141
13	OliveOil	0.133	0.233	0.137	0.133	0.479	0.212	0.141
14	SwedishLeaf	0.210	0.173	0.169	0.128	0.211	0.254	0.106
15	Trace	0	0	0	0	0.24	0	0.04
16	Two_Patterns	0	0	0.002	0	0.09	0.052	0.001
17	Synthetic_Control	0.007	0.020	0.012	0.070	0.120	0.150	0.060
18	Wafer	0.02	0.031	0.007	0.004	0.005	0.018	0.005
19	50Words	0.310	0.314	0.249	0.233	0.369	0.341	0.251
20	Yoga	0.164	0.161	0.149	0.145	0.170	0.130	0.123
	Aver.	0.179	0.182	**0.169**	**0.153**	0.211	0.197	0.192

one time series in the testing set to search the most similar object in the training set. It means that every time series in the testing set is regarded as the query series and the most similar object searched from the training set are regarded as the queried one. Then we compare that whether the labels of the query and the queried are the same. If the labels are equal, we consider it as the right classification. Otherwise, it is a wrong classification.

In the classification, two distance functions, DTW based on Sakoe-Chiba Band (SDTW) and the proposed one based incremental warping window (IDTW) are used to measure the similarity between the query series and the queried one. In addition, the original DTW is used to compare with the two methods so as to show the improvement of the warping operation. After completing the classification for each time series in the testing dataset, we record the number of the wrong classifications and can obtained the error rate. In the two method IDTW and SDTW, the percentage r is set to 0.1. The original DTW need not require no warping window. In the proposed method IDTW, the beginning length of the incremental warping window is set to from 0 to 9, that is $b = \{0, 1, \cdots, 9\}$. We average all the classification results according to different value of b as the final result. In addition, some other traditional methods such as Euclidean distance *ED*, *SpADe* [44] and $2DD_{LCSS}$ [45] are used to compare their classification performance. The classification results are produced by the six methods are shown in Table 2.

Compared with the classification results of DTW, SDTW and IDTW as shown in Table 2, we know that in most cases the mean of error rates and the minimal results of IDTW are often less than that of SDTW and DTW. In some cases, SDTW only outperforms IDTW in five datasets such as Adiac, CBF, FaceAll, FaceFour and Two_Patterns. It means that the proposed method IDTW is better to measure the similarity between time series in most of the above mentioned datasets than SDTW. However, it is easy to find that the

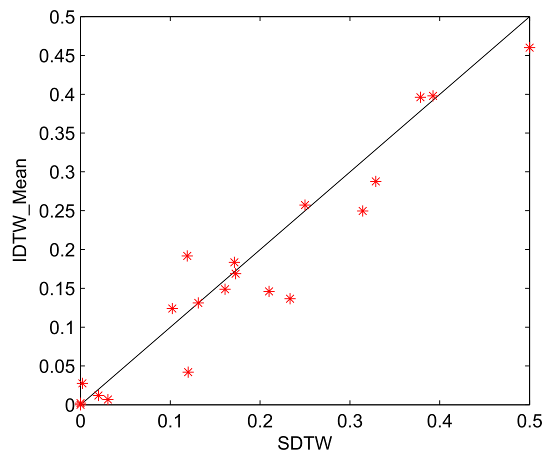

FIGURE 6. The classification comparisons between IDTW and IDTW_Mean.

methods SDTW and IDTW based on warping window can obtain better classification results.

Since this work focuses on the improvement of the DTW based on the warping cost matrix and the classification results of the DTW based methods (like the original DTW, IDTW and SDTW) are better than the other three ones (ED, SpADe and $2DD_{LCSS}$), we further visualize the classification results for the DTW-based methods. Namely, SDTW is respectively compared with IDTW_Mean and IDTW_Min as shown in Figure.6 and Figure.7, which indicates that the proposed methods IDTW can obtain better classification results than SDTW.

C. EFFICIENCY COMPARISON

Beside the accuracy of the similarity measure between two time series using the warping methods, the efficiency is also important. Since the original DTW without warping window must compute the best warping path in the whole cost matrix, which costs the highest time in the three methods.

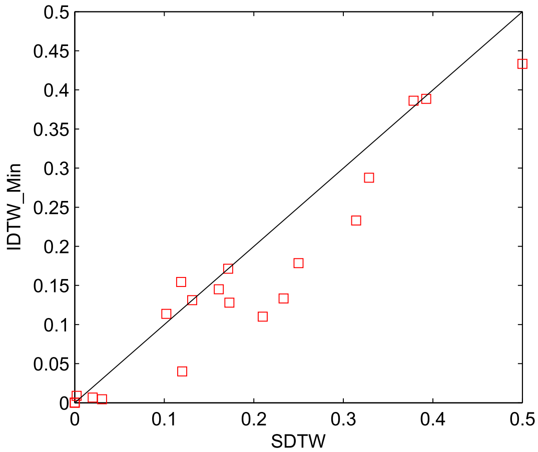

FIGURE 7. The classification comparisons between IDTW and IDTW_Min.

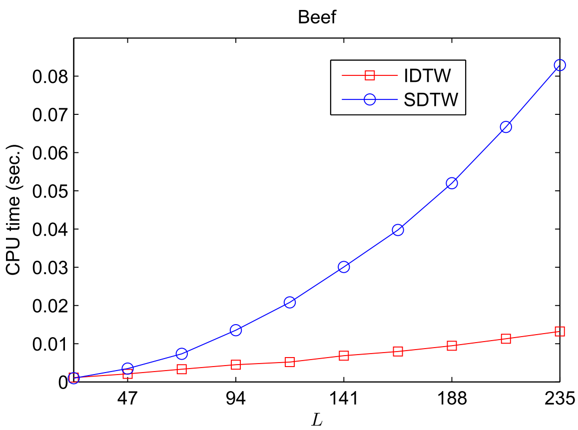

FIGURE 8. Time consumption of two methods SDTW and IDTW in the Beef according to different lengths of compared subsequences.

In this section, we only compare the time consumption of the two methods SDTW and IDTW based on warping window. They are performed on two datasets that randomly chosen from Table 1. Each pair of time series are equally segmented into 10 subsequences that are used to simulate data stream. It means that the lengths of the obtained subsequences from time series in the two datasets are respectively {47, 94, 141, 188, 235, 282, 329, 376, 423, 470} and {57, 114, 171, 228, 285, 342, 399, 456, 513, 570}. Let SDTW and IDTW compute the similarity between two subsequences with the same lengths. Since IDTW is an incremental warping method, it can use the previous cost vectors to rebuilt the new ones, which will speed up the whole calculation.

In this experiment, let every one subsequence of time series in the testing set compare all the same-length subsequences of time series in the training set. We record time cost of every comparison and average them as the last results according to different lengths. The time comparisons of two time series datasets are respectively shown in Figure.8 and Figure.9.

It is easy to find that the time consumption of the proposed method IDTW is much less than that of SDTW in terms of different lengths of the compared subsequences. It means that

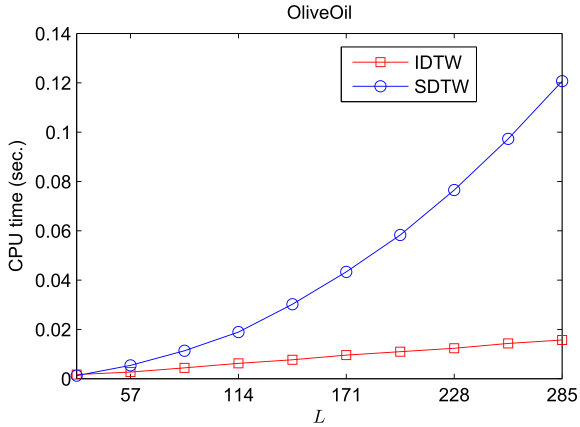

FIGURE 9. Time consumption of two methods SDTW and IDTW in the OliveOil according to different lengths of compared subsequences.

the proposed method IDTW is more efficient than SDTW. Moreover, IDTW is an incremental method to compute the distance between two time series data stream.

V. CONCLUSIONS

Due to the disadvantage and importance of dynamic time warping based on Sakeo-Chiba Band, we propose an improved version to speed up the calculation of the optimal warping path in the reduced scope of cost matrix. In this work, the reduced scope like Sakoe-Chiba Band are constructed by an incremental warping window, which can reflect more relationship about the information of the recent data points. It makes the proposed method more suitable for similarity measure between any two stream time series. There are three advantages as follows:

(1) The incremental warping window instead of the traditional constraints on the scope of the cost matrix can obtain less number of the cells that encloses the optimal warping path. It means that the proposed method is more faster than the traditional one.

(2) Since the length of the incremental warping window is changeable with the length of the compared time series, the proposed method based on the incremental warping window can be used to measure the similarity between stream time series. It extends DTW to measure the similarity between any two stream time series.

(3) DTW based on the incremental warping window concentrates on much more information of the recent data points rather than the previous ones, which reflects that the nearer the data points are the more important their information is.

As far as we known, DTW based on Sakeo-Chiba Band is very important and often extended to speed up the calculation of similarity search and index for time series. One of the popular methods is LB_Keogh [35], which has already been applied to many domains. One of the future work is to design a lower bound function on IDTW like LB_Keogh to fast remove the dissimilar objects in the time series datasets.

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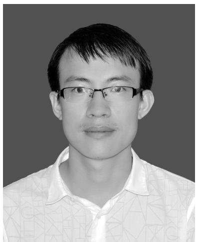

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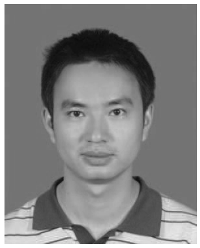

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